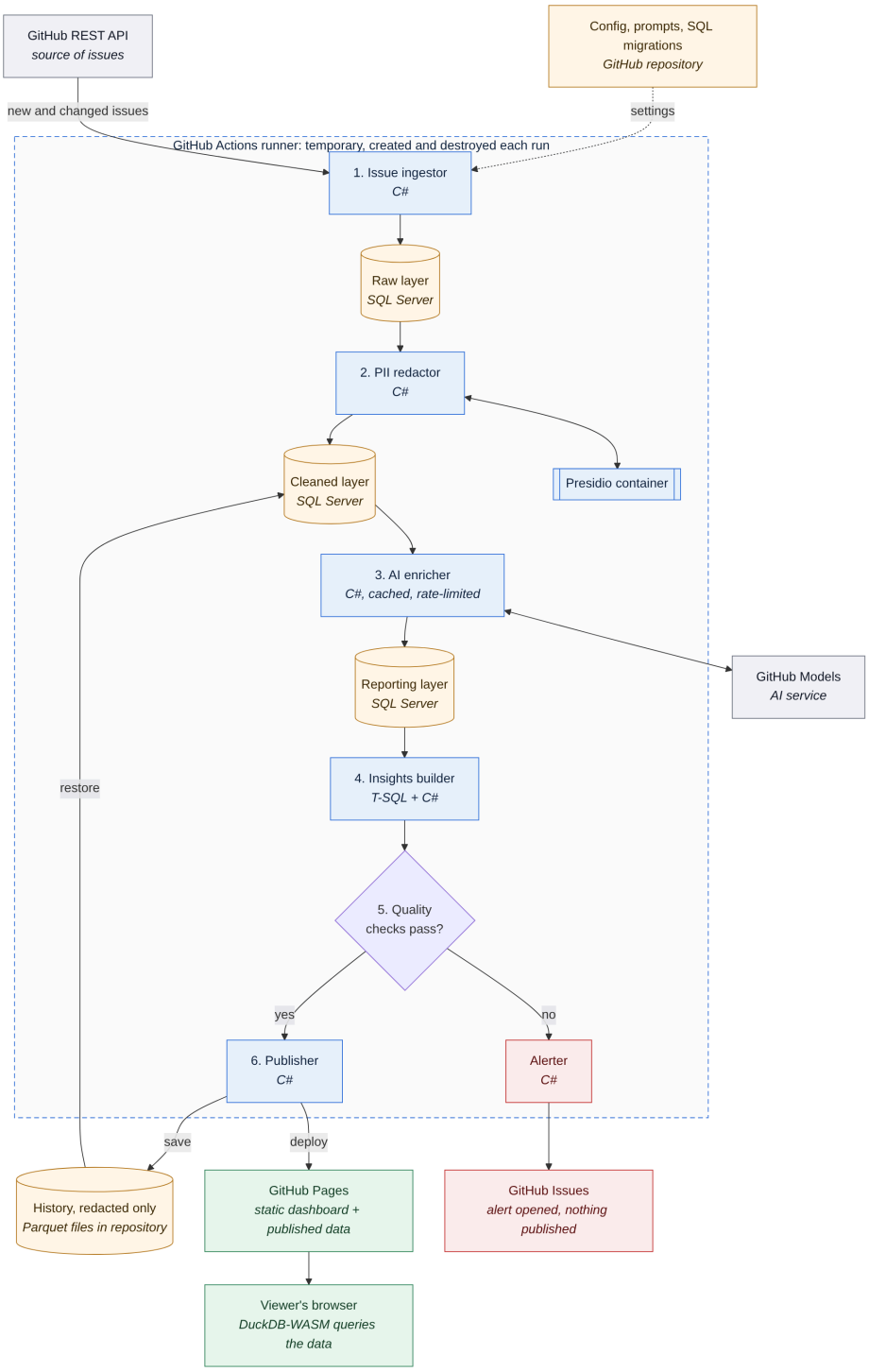
Design: Feedback Intelligence Platform

Status	Approved
Version	1.5
Date	2026-11-01
Owner	Benjamin Rose
Related	Project plan, Requirements, Data dictionary, Decision records

This document describes how the Feedback Intelligence Platform is built: its architecture, components, failure handling, configuration, dashboard, security, and repository layout. Detailed table definitions are in the data dictionary, and the reasoning behind each major choice is in the decision records.

1. Architecture

A scheduled GitHub Actions workflow runs a C# pipeline once a day. Each run starts temporary SQL Server and Presidio containers, restores history, processes new and changed issues through raw, cleaned, and reporting layers, and publishes a static dashboard to GitHub Pages. The runner is destroyed afterward, so nothing survives except what is deliberately saved.



Source: `diagrams/architecture.mmd` *(Mermaid).*

Environment	What runs there
GitHub Actions runner (temporary)	The C# pipeline, SQL Server Express, and Presidio containers
GitHub repository	Code, config, prompts, migrations, docs; history on the `data` branch
GitHub Pages	The static dashboard and its published data files

Environment	What runs there
Viewer's browser	Dashboard code and DuckDB-WASM
External services	GitHub REST API (issues) and GitHub Models (AI)

Key decisions: ADR 0002 (GitHub as platform), 0004 (temporary SQL Server), 0005 and 0013 (history), 0009 (data layers).

2. Data design

Data moves through four SQL Server schemas:

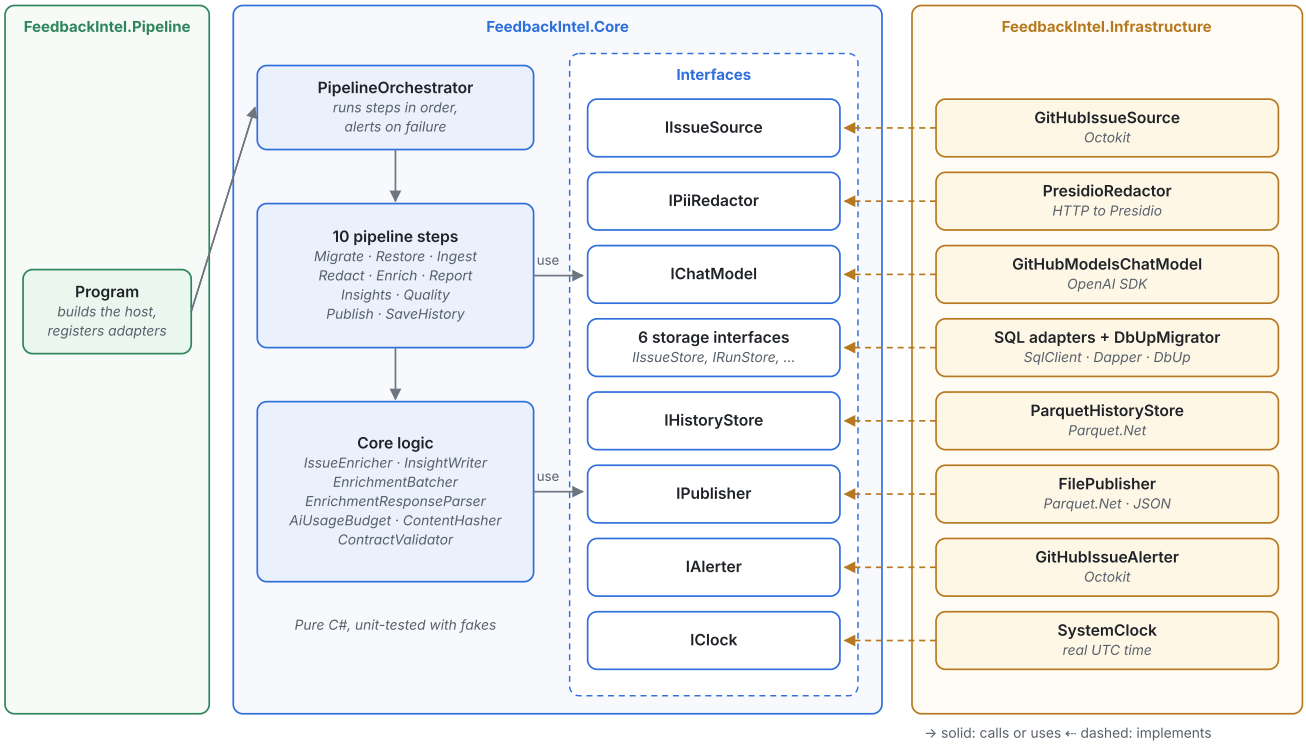
Schema	Holds	Saved as history	Published
`raw`	Issues as fetched, unredacted	Never	Never
`cleaned`	Redacted issues, AI results, digests	Yes (without issue bodies)	No
`reporting`	Aggregates and dashboard tables, rebuilt each run	No	Yes
`ops`	Run records and watermarks	Yes	Run health only

The **data contract** lists every field allowed in published and history files; an automated test enforces it. Full definitions: data dictionary.

3. Components

Infrastructure implements Core's interfaces. Core depends on nothing.

Dependencies point inward: Pipeline → Infrastructure → Core. Swapping a service means replacing one adapter.



Project	Responsibility
`FeedbackIntel.Core`	Data models, interfaces, pipeline steps, and pure logic (batching, validation, hashing, data contract). No external dependencies, so it is fully unit-testable with fakes.
`FeedbackIntel.Infrastructure`	Adapters that implement Core's interfaces using real services: Octokit, Presidio over HTTP, the OpenAI SDK for GitHub Models, SqlClient with Dapper and DbUp, and Parquet.Net.
`FeedbackIntel.Pipeline`	Console entry point. Loads configuration, registers adapters through dependency injection, and starts the orchestrator.
`FeedbackIntel.Core.Tests`	Fast unit tests using fakes.
`FeedbackIntel.Integration.Tests`	Tests against real SQL Server and Presidio containers via Testcontainers.
`FeedbackIntel.Evaluation`	Scores prompts against the labeled evaluation set (FR-DEV-02).

Dependency rule: Infrastructure depends on Core; Core depends on nothing. Every external service sits behind an interface, so it can be faked in tests or replaced by another provider (NFR-MNT-03). Class-level documentation lives in the code as XML doc comments.

Interfaces: `IPipelineStep`, `IIssueSource`, `IPiiRedactor`, `IChatModel`, `IIssueEnricher`, `IInsightWriter`, `IDatabaseMigrator`, `IIssueStore`, `ICleanedIssueStore`, `IWatermarkStore`,

`IEnrichmentStore`, `IRunStore`, `IReportingBuilder`, `IQualityChecker`, `IHistoryStore`, `IPublisher`, `IStaleDataReporter`, `IClock`.

4. Pipeline flow and error handling

Rules

1. **Temporary errors are retried; permanent errors are not.** Network errors, timeouts, and rate-limit responses are retried up to 3 times with increasing waits (2, 4, 8 seconds). Bad data, bad configuration, and code errors fail immediately.
2. **Any failure stops the run.** The orchestrator records the failed step and error, saves the run record, and exits with a failure code; the workflow's alert job then raises the alert (see *Alerts* below). Nothing later runs, so nothing half-finished is published.
3. **A failed run is never harmful.** The dashboard keeps the last good data, and the watermark doesn't move, so the next run picks up everything missed.
4. **Timeouts:** the pipeline step has a 40-minute limit inside a 50-minute job, leaving time for the failure-alert step. The normal run target is 30 minutes (NFR-PERF-03).
5. **Logging:** every log entry carries the run ID and step name automatically (NFR-OBS-01). Logs never include issue text, because run logs of a public repository are public.

Steps

#	Step	On failure	Retried?
—	Freshness check (after restore)	Stale-data alert if the last successful run finished more than 48 hours ago; the run continues	—
1	Migrate database	Alert	Waits up to 2 minutes for SQL Server; script errors not retried
2	Restore history	Alert, unless it is a manual **bootstrap** run	No
3	Ingest issues	Alert; also fails if more than 10% of records are rejected, once at least 20 were fetched	Yes; waits for a rate-limit reset only if under 15 minutes
4	Redact	Alert; **never continues without redaction**	Yes
5	Enrich	Not a run failure: invalid answers become `unclassified`; unprocessed issues stay `pending`	Yes; requests spaced about 4 seconds apart; every attempt counts against the daily budget
6	Rebuild reporting	Alert	No
7	Build insights	Not a run failure: topics show "unavailable today"; a missing digest is generated on the next run	Yes

#	Step	On failure	Retried?
8	Quality checks: row counts, required fields, valid values, freshness	Alert listing every failed check and how many rows failed it; nothing published	No
9	Publish (write files)	Alert; contract violations are treated as possible privacy issues	Write errors yes; violations no
10	Advance watermarks	Alert. Runs only after publishing succeeded, so a failed run never skips issues	No
11	Save history (write files)	Alert	Yes
—	Workflow: deploy Pages, push history	Alert from the same alert job	Yes

Two deliberate exceptions: AI problems never fail the run, because the AI is an enhancement and the dashboard should still update. Redaction problems always fail it, because privacy is never traded for availability.

Alerts

The pipeline doesn't write issues itself. It passes what it found to the workflow as step outputs (failed step, error, and data age when stale), and a separate **alert** job, the only part of the system allowed to write issues (NFR-SEC-02), acts on them. The alert job runs after every run, whatever happened, so a build error, timeout, or failed deploy is reported too (NFR-OBS-02).

Situation	What the alert job does
Any job failed or was cancelled	Comments on the open alert issue, or opens one if none is open (FR-OPS-02, FR-OPS-03). The issue has the failed job and step, the error, and the run link.
Data was stale (older than `StaleDataAlertHours`)	Adds a stale-data note to the failure alert; if the run succeeded and nothing was open, opens an alert and closes it at once, so the gap is on record and notified (NFR-REL-02).
The run succeeded and an alert is open	Comments that it is resolved, with the run link, and closes it.

At most one alert issue, labeled `pipeline-alert`, is open at a time; later problems become comments on it. Error text passes through two filters first: SQL Server errors that can quote data values (such as truncation and conversion errors) are reduced to their error number, and the text is cut to one line of at most 500 characters. Values reach the alert script only through environment variables, so error text can't run as a command.

Backfills

A manual run can take a backfill date range (FR-ING-06, US-14). Besides the normal incremental fetch, it re-fetches every issue *created* in that range, so those issues are redacted again and upserted without duplicates (FR-DAT-03), for example after a redaction or prompt improvement. Both fetches happen in one

request stream: GitHub only filters by update time, and every issue created in the range was updated on or after its creation date, so fetching from the earlier of the range start and the watermark finds both sets. The watermark stays correct, because a re-fetched issue that wasn't updated since the watermark is older than it. A range may start at most 365 days ago, and the run record keeps the range in `backfill_from` and `backfill_to`.

5. Configuration

All tunable behavior lives in `config/pipeline.json` (FR-CFG-01). The settings classes in Core are its schema: at startup every value is checked against its allowed range, and in CI a test reads the real file with the same code, also rejecting any setting the code doesn't know, such as a misspelled name, so a bad config change fails its pull request (FR-CFG-02). Any value can be overridden by an environment variable, which is how local runs redirect output away from the live site.

```
{
  "Repositories": [
    { "Owner": "microsoft", "Name": "winget-cli", "Enabled": true },
    { "Owner": "microsoft", "Name": "PowerToys",  "Enabled": false },
    { "Owner": "microsoft", "Name": "vscode",     "Enabled": false }
  ],
  "Ingestion":      { "InitialHistoryDays": 90, "MaxRejectedPercent": 10,
"MinIssuesForRejectionCheck": 20,
                     "RateLimitMaxWaitMinutes": 15 },
  "Classification": { "Categories": [ "bug", "feature_request", "question", "docs", "performance"
],
                     "Severities": [ "low", "medium", "high", "critical" ] },
  "Ai":             { "Endpoint": "https://models.github.ai/inference", "Model": "openai/gpt-4.1-
mini",
                     "PromptDirectory": "prompts/v1", "DailyRequestLimit": 140,
                     "MinSecondsBetweenRequests": 4, "MaxIssuesPerRequest": 10,
                     "MaxCharactersPerRequest": 24000, "MaxRetries": 3 },
  "Text":           { "BodyMaxCharacters": 4000, "SummaryMinBodyCharacters": 500 },
  "Insights":       { "TopicMinIssues": 3, "TopicWindowDays": 7, "DigestMaxWords": 300 },
  "Operations":     { "StaleDataAlertHours": 48, "SqlStartupWaitSeconds": 120 },
  "Privacy":        { "AnalyzerUrl": "http://localhost:5002", "AnonymizerUrl":
"http://localhost:5001",
                     "ScoreThreshold": 0.5 },
  "Publishing":     { "OutputDirectory": "out/site", "HistoryDirectory": "out/history" }
}
```

Not in this file	Where it lives	Why
Schedule and timeouts	Workflow file	GitHub reads these only from workflow files
Backfill date range (`Backfill:From` , `Backfill:To`)	Manual workflow inputs, passed as environment variables	It applies to a single run, not every run
Tokens and passwords	Environment variables	Secrets never enter the repository (NFR-SEC-01)

Not in this file	Where it lives	Why
Database connection string	Environment variable	Differs between local, CI, and production

Notes:

- **Categories and severities come from config, not code enums,** so they can change through a pull request (US-12). Because changing them also affects the prompts and evaluation labels, CI checks that all three agree.
- **The model name is an example,** to be confirmed in M4 against the evaluation set.
- **The daily AI limit is 140,** deliberately below GitHub's limit of about 150 to leave a safety margin.

6. Dashboard

A static site of plain HTML and JavaScript, with charts drawn by Chart.js and queries run in the browser by DuckDB-WASM (ADR 0008).

Page	Shows	Reads	Milestone
Overview	Last 7 days vs. previous 7: new, closed, % bugs, critical, median days to close	`site_info.json`, `overview.json`	M3
Trends	Weekly volume, closures, and category mix; repository and date filters	`weekly_metrics.parquet`, `weekly_category.parquet`	M3
Issue explorer	Searchable, filterable issue table with links to GitHub; CSV export (M8)	`issues.parquet`	M3
Emerging topics	Ranked fast-growing topics; select one to list its issues	`topics.parquet`, `topic_issues.parquet`, `issues.parquet`	M5
Weekly digest	Latest digest and archive	`digests.parquet`	M5
Pipeline health	Last 30 runs, data freshness, AI usage vs. limit	`run_health.parquet`, `site_info.json`	M5

Loading: the Overview page renders immediately from two small JSON files; DuckDB-WASM loads in the background and is needed only for the other pages. This addresses the 3-second load target (NFR-PERF-02).

Every page shows: a repository selector, the time the data was last updated (FR-DSH-05), and an "AI-generated" marker next to AI categories, severities, summaries, topics, and digests (FR-DSH-06).

Accessibility: responsive layout for desktop and mobile (NFR-USE-01); WCAG AA contrast, and charts that never rely on color alone (NFR-USE-02).

7. Security and privacy

Threat	Mitigation
Personal information in issue text reaches the AI, history, or dashboard	Presidio redaction before anything leaves the raw layer; usernames never collected; bodies never saved or published; data contract test; PII test set in CI (FR-PRV-01 to 04, ADR 0013)
Issue text printed in public run logs	Logs record counts, IDs, and step names only, never issue text. Run logs of a public repository are visible to anyone.
Malicious HTML or script in issue titles or AI output (cross-site scripting)	The dashboard inserts all data as plain text, never as HTML, and the site sets a Content Security Policy that blocks inline scripts.
Prompt injection through issue text	Issue text is clearly marked as data in prompts; AI output is parsed as JSON and validated against configured values; the AI has no ability to act (NFR-SEC-03, risk R4)
Leaked credentials	Only the built-in, short-lived `GITHUB_TOKEN` is used; secret scanning in CI (NFR-SEC-01)
Over-privileged workflow	Each job declares minimum permissions; only the alert step can write issues, and only the deploy step can write Pages (NFR-SEC-02)
Compromised dependency or third-party action	Dependency vulnerability scanning in CI; third-party GitHub Actions pinned to exact commit versions (NFR-SEC-04)
Incorrect or harmful data published	Quality checks gate publishing; the contract validator blocks unexpected fields; failures publish nothing

8. Repository structure

```
feedback-intelligence/
├── README.md                    ← front door: what it is, live link, how to run
├── CONTRIBUTING.md              ← workflow, standards, AI-assisted development guide
├── .gitattributes               ← consistent line endings (ADR 0012)
├── .github/
│   ├── workflows/               ← ci.yml, pipeline.yml, evaluate.yml
│   └── pull_request_template.md ← self-review checklist and AI disclosure
├── config/                      ← pipeline.json and its schema
├── prompts/v1/                  ← versioned AI prompts
├── database/migrations/         ← numbered .sql scripts (DbUp)
├── src/
│   ├── FeedbackIntel.Core/
│   ├── FeedbackIntel.Infrastructure/
│   └── FeedbackIntel.Pipeline/
├── tests/
│   ├── FeedbackIntel.Core.Tests/
│   ├── FeedbackIntel.Integration.Tests/
│   └── FeedbackIntel.Evaluation/   ← includes the labeled evaluation set
├── dashboard/                   ← static site source
├── deploy/                      ← Bicep templates, validated in CI only (M7)
├── docker-compose.yml           ← local SQL Server and Presidio
└── docs/                        ← plan, requirements, design, data dictionary, ADRs, diagrams
```

The `data` branch holds only history files and is written only by the pipeline workflow.

9. Open items to verify during the build

Item	When	Status
Whether daily commits to the `data` branch keep scheduled workflows active (ADR 0005, risk R6)	M3	Moved from M2, because the `data` branch arrives with history in M3

Item	When	Status
SQL Server container startup time on GitHub runners (assumption A4)	M2	Done: the container starts during the build step and is ready before the pipeline needs it (migration 0.7 s; whole run 49 s)
Presidio accuracy on real issues, including custom recognizers (assumption A5)	M3	Open
Dashboard load time with DuckDB-WASM (NFR-PERF-02)	M3	Open
Model choice and prompt accuracy against the evaluation set (assumption A6)	M4	Open

Revision history

Version	Date	Change
1.5	2026-11-01	Added *Backfills*. Config validation in CI now reads the real file with the pipeline's own settings classes and rejects unknown settings, instead of a separate JSON schema that could drift from the code.
1.4	2026-11-01	Added the *Alerts* section: alerts are raised by a separate workflow job from the pipeline's step outputs. Listed the quality checks; replaced `IAlerter` with `IStaleDataReporter`, and added `ICleanedIssueStore` and `IWatermarkStore` to the interface list.
1.3	2026-11-01	Added the advance-watermarks step before saving history; the rejection limit applies once at least 20 issues were fetched, so small incremental runs aren't failed by one unusual issue.
1.2	2026-10-31	Added the `Privacy` section to the configuration sample.
1.1	2026-10-24	Recorded results of the M2 open items.
1.0	2026-10-18	Initial design approved